

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0633

Yauco Plaza 138 kV
May 30, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0633. The Interconnection Customer proposes a 221.6 MW wind facility interconnecting at the Yauco Plaza 138 kV substation in the Tennessee area. The request was received on June 18, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 219 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$46,294,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Clinch River - Carlton Hills 345kV, the Hecata - S. Lexington-Duxbar 138kV L loads to 164.1% of its 130 MVA rating. Project DFAX: 100.0%.

S Sheboygan Falls - Sunny Slope Substation 138kV line: 163.8% loading (211 MVA) following the outage of the Baker - Hubbardstown 345kV; DFAX = 100.0%.

The N City - Sunny Slope 138kV is overloaded to 163.6% of its 200 MVA emergency rating for the loss of the Wixom - Albion Road 138kV Line. The project contributes 100.0% DFAX (221.6 MW) to this overload.

The Greenville Ave - Beanblossom-Sw Sta 138kV is overloaded to 157.4% of its 284 MVA emergency rating for the loss of the Livermore - Lee County Solid Waste Energy Substation 345kV. The project contributes 19.3% DFAX (42.7 MW) to this overload.

Sand Springs - Snohomish 138kV: 154.2% loading (262 MVA) following the outage of the Sand Springs - Ojo 138kV; DFAX = 8.1%.

The Wixom - Albion Road Substation 138kV line is overloaded to 152.3% of its 593 MVA emergency rating for the loss of the N Appleton - Davisville 138kV L. The project contributes 9.1% DFAX (20.3 MW) to this overload.

The Texaco Bt - Collins 138kV is overloaded to 147.8% of its 173 MVA emergency rating for the loss of the Wixom - Albion Rd 138kV. The project contributes 5.4% DFAX (12.0 MW) to this overload.

The Mistersky - Guadalupe 138kV L is overloaded to 146.3% of its 174 MVA emergency rating for the loss of the Wissota - Berry Street 138kV. The project contributes 8.5% DFAX (18.8 MW) to this overload.

For the loss of Snohomish - Sand Springs 138kV Line, the Ojo Substation - Sand Springs 138kV loads to 140.3% of its 304 MVA rating. Project DFAX: 7.8%.

The Anson N - Cheoah 138kV L is overloaded to 139.8% of its 247 MVA emergency rating for the loss of the Wixom - Albion Road 138kV. The project contributes 5.4% DFAX (12.0 MW) to this overload.

The Texaco T - Letterkenny 138kV L is overloaded to 139.6% of its 195 MVA emergency rating for the loss of the Albion Rd - Wixom Substation 138kV. The project contributes 5.4% DFAX (12.0 MW) to this overload.

For the loss of Wixom - Albion Road 138kV, the Cheoah - Summit Ave 138kV loads to 138.7% of its 235 MVA rating. Project DFAX: 5.4%.

Sonoita - Summit Ave 138kV: 137.9% loading (231 MVA) following the outage of the Wixom - Albion Road 138kV line; DFAX = 5.4%.

The Del Norte - Sonoia 138kV is overloaded to 137.3% of its 228 MVA emergency rating for the loss of the Wixom - Albion Road 138kV. The project contributes 5.4% DFAX (12.0 MW) to this overload.

For the loss of Wixom - Albion Rd 138kV Line, the Mueller - Letterkenny 138kV line loads to 135.9% of its 216 MVA rating. Project DFAX: 5.4%.

For the loss of Berry T - Wissota 138kV, the Guadalupe - Mortan 138kV loads to 135.3% of its 201 MVA rating. Project DFAX: 8.5%.

For the loss of Albion Road - Wixom 138kV line, the Del Norte - Mueller 138kV loads to 135.3% of its 224 MVA rating. Project DFAX: 5.4%.

The Mortan - Mainwring 138kV is overloaded to 134.4% of its 204 MVA emergency rating for the loss of the Wissota - Bery St 138kV. The project contributes 8.5% DFAX (18.8 MW) to this overload.

Claus - San Andreas 138kV: 132.1% loading (249 MVA) following the outage of the Davisville - N Appleton 138kV; DFAX = 9.7%.

The Union - Claus 138kV is overloaded to 131.5% of its 236 MVA emergency rating for the loss of the Davisville - North Appleton 138kV. The project contributes 9.7% DFAX (21.5 MW) to this overload.

The Wissota - San Andreas 138kV is overloaded to 131.4% of its 253 MVA emergency rating for the loss of the Davisville - North Appleton 138kV. The project contributes 9.7% DFAX (21.5 MW) to this overload.

The Surfside - Amargosa 138kV is overloaded to 130.6% of its 269 MVA emergency rating for the loss of the Livermore - Lee County Solid Waste Energy Substation 345kV. The project contributes 19.6% DFAX (43.4 MW) to this overload.

For the loss of Livermore - Lee County Solid Waste Energy 345kV L, the Amargosa - Dewitt Substation 138kV loads to 128.2% of its 267 MVA rating. Project DFAX: 19.6%.

For the loss of Berry Street - Northfield Substation 345kV, the Loveland - Ojo 138kV loads to 127.0% of its 223 MVA rating. Project DFAX: 7.6%.

The Balta - S. Lexington-Duxbar 138kV is overloaded to 125.1% of its 172 MVA emergency rating for the loss of the Burgess Biopower - Yellowcake 138kV. The project contributes 100.0% DFAX (221.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S. Lexington-Duxbar - Hecata 138kV	Rebuild line with higher-capacity conductor	\$6,771,000	100% (\$6,771,000)
S Sheboygan Falls - Sunny Slope Substation 138kV line	Rebuild line with higher-capacity conductor	\$8,465,000	100% (\$8,465,000)
North City - Sunny Slope 138kV	Rebuild line with higher-capacity conductor	\$2,192,000	100% (\$2,192,000)
Greenville Ave - Beanblossom-Sw Sta 138kV line	Rebuild line with higher-capacity conductor	\$29,166,000	11% (\$3,328,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sand Sprins - Snohomish 138kV	Rebuild line with higher-capacity conductor	\$21,415,000	5% (\$1,047,000)
Wixom - Albion Road Substation 138kV line	Rebuild line with higher-capacity conductor	\$2,179,000	2% (\$55,000)
Texaco Bt - Collins 138kV	Rebuild line with higher-capacity conductor	\$6,767,000	5% (\$332,000)
Mistersky - Guadalupe 138kV L	Rebuild line with higher-capacity conductor	\$10,180,000	6% (\$621,000)
Ojo Substation - Sand Springs 138kV	Rebuild line with higher-capacity conductor	\$9,550,000	4% (\$414,000)
Anson N - Cheoah 138kV L	Rebuild line with higher-capacity conductor	\$3,110,000	5% (\$153,000)
Texaco T - Letterkenny 138kV L	Rebuild line with higher-capacity conductor	\$15,447,000	5% (\$759,000)
Summit Ave - Cheoah 138kV	Rebuild line with higher-capacity conductor	\$4,930,000	5% (\$242,000)
Sonoita - Summit Ave 138kV	Rebuild line with higher-capacity conductor	\$6,600,000	5% (\$324,000)
Del Norte - Sonoita 138kV L	Rebuild line with higher-capacity conductor	\$5,364,000	5% (\$263,000)
Mueller - Letterkenny 138kV	Rebuild line with higher-capacity conductor	\$3,948,000	5% (\$194,000)
Guadalupe - Mortan 138kV	Rebuild line with higher-capacity conductor	\$20,649,000	6% (\$1,259,000)
Del Norte - Mueller 138kV	Rebuild line with higher-capacity conductor	\$16,818,000	5% (\$826,000)
Mortan - Mainwring 138kV	Rebuild line with higher-capacity conductor	\$12,347,000	6% (\$753,000)
Claus - San Andreas 138kV	Rebuild line with higher-capacity conductor	\$5,908,000	6% (\$382,000)
Union - Claus 138kV	Rebuild line with higher-capacity conductor	\$1,855,000	6% (\$120,000)
Wissota - San Andreas 138kV	Rebuild line with higher-capacity conductor	\$8,452,000	6% (\$547,000)
Surfside - Amargosa 138kV	Rebuild line with higher-capacity conductor	\$7,842,000	22% (\$1,742,000)
Amargosa - Dewitt Substation 138kV	Rebuild line with higher-capacity conductor	\$6,762,000	22% (\$1,503,000)
Loveland - Ojo 138kV	Rebuild line with higher-capacity conductor	\$4,909,000	6% (\$281,000)
Balta - S. Lexington-Duxbar 138kV	Rebuild line with higher-capacity conductor	\$13,721,000	100% (\$13,721,000)
	Total allocated to project		\$46,294,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.